

Avi Guha

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EDUCATION

University of British Columbia

B.A.Sc. Engineering Physics — Electrical and Computer Specialization

Vancouver, BC

Sept 2023 — May 2028

- Cumulative GPA: 3.7
- Relevant Courses: Data Structures & Algorithms, Circuit Design, Digital Systems and Microcomputers, Machine Learning.

WORK EXPERIENCE

Quantum Silicon Photonics Researcher

Jan 2026 — Present

Stewart Blusson Quantum Matter Institute

Vancouver, BC

- Researched the use of superconducting nanowire photon detectors on **silicon** for photon-based quantum computers.
- Engineered a variable low noise current source using a **DAC**, controlled with **SPI**, to reliably bias superconducting wires.
- Implemented a low-noise amplifier with a 50 MHz - 3 GHz bandwidth and less than 100 mW power dissipation.
- Developed a high power RF amplifier using **Cadence Allegro** to maximize the signal to noise ratio from photon detectors.

Electrical and Firmware Engineer

Sept 2024 — Present

UBC Thunderbots

Vancouver, BC

- Led the design of a power distribution board which galvanically isolated low (3.3V - 24V) and high (240V) voltages.
- Engineered linear, switching, and flyback voltage regulators to precisely distribute power to components across the robot.
- Designed an **SPI-CAN** interface board, mitigating EMI effects, improving signal integrity on critical communication lines.
- Implemented trans-impedance amplification based current sensing, enabling field oriented control of three-phase motors.
- Optimized the layout of a three-phase motor driver board, improving gate driver stability and current sense accuracy.
- Spearheaded CI firmware deployment by interfacing a **Raspberry PI** and multiple STM32s over **SWD** enabling autoflash.
- Developed firmware in **C** for an **STM32**, implementing real-time **FOC** algorithms for **20%** more efficient motor control.

2D Physics Research Intern

Jan 2025 — May 2025

University of Calgary

Calgary, AB

- Researched the tribological properties of 2D materials for applications in **MEMS** / **NEMS** using atomic force microscopy.
- Developed a **PID** controlled fluid cell, enabling researchers to examine the effect of humidity on 2D materials.
- Developed a full-cycle **Java** application using **OpenCV**, allowing researchers to visualize wear due to repetitive forces.
- Designed a TEM sample holder in **Solidworks** with 2 μm tolerances that operated in ultra-high vacuum environments.

PROJECTS

Autonomous Pet Rescuing Robot

github.com/avi-guha/ROBOT

- Worked within a team of four to design and build a robot capable of rescuing pets for a competition.
- Designed a dual H-bridge inverter on a PCB to drive two DC motors, incorporating gate drivers and flyback protection.
- Spearheaded the design of an **ESP-32** / **RasPI** GPIO board to autonomously control stepper motors and line following.
- Implemented **UART** communication between an ESP-32 and RasPI, enabling computer vision and hardware integration.
- Utilized **FreeRTOS** to implement dual-core line following on an ESP-32 using digital photoelectronic signals.
- Developed firmware to control stepper motors motion with an S-curve velocity profile, eliminating skipped steps.
- Trained a **YoloV11** computer vision model capable of running on a RasPI that recognized pets in diverse environments.
 - Reached 95% validation accuracy and <2 degree variance when estimating central angles.

Virtual Autonomous Driving Neural Network

github.com/avi-guha/ENPH-353-COMPETITION

- Designed neural networks for a **ROS** simulation, enabling a vehicle to autonomously drive through various environments.
- Developed a convolutional neural network in **Tensorflow** to accurately read words and sentences on road signs.
- Trained a **deep neural network** in **PyTorch** to autonomously drive using **imitation learning** and data augmentation.
- Designed a multimodal CNN-MLP architecture, combining camera-based features and LiDAR readings for vehicle control.

SKILLS

- **Programming:** Java, Python, C/C++, Rust, Git, Unix, Linux, FreeRTOS, PyTorch, Tensorflow, Matlab, ZephyrRTOS
- **Engineering:** Verilog, VHDL, Debugging, Altium, Cadence Virtuoso & Allegro, Robotics, Computer Architecture